

# Pulling the Right Training Levers: Randomized Optimization, Adam-Family Ablations, and Regularization on the Covertypes PyTorch Backbone

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**Abstract**—Using the PyTorch MLP backbone from the SL Report as a fixed Covertypes backbone ( $n=20,000$ , seed 7641, 60/20/20 stratified split, `StandardScaler` fit on train only), this report interrogates three training forces under controlled compute budgets. Part 1 applies Randomized Hill Climbing (RHC), Simulated Annealing (SA), and a Genetic Algorithm (GA) to the final two layers (1,159 trainable parameters, well within the 50k cap); Part 2 runs seven Adam-family optimizer ablations — plain SGD, SGD+momentum, Nesterov, Adam, Adam without bias correction, Adam with  $\beta_1=0$  (RMSProp-like), and AdamW — under budget-matched gradient evaluations with sensitivity heatmaps and three-seed stability bands; Part 3 studies five regularizers under fixed best-Part-2 Adam settings: L2 weight decay, early stopping, dropout, label smoothing, and input noise; Part 4 (extra credit) composes the best pieces from Parts 1–3. Hypotheses: Adam-family optimizers will reach a validation-loss threshold faster than plain SGD because adaptive second-moment scaling reduces sensitivity to Covertypes’s heterogeneous gradient scales [8], but this speed advantage will not uniformly transfer to Macro-F1 gains on imbalanced minority classes; and regularization, not optimizer choice, will primarily drive minority-class recall improvements. Adam without bias correction converges faster in the first 10 epochs then matches standard Adam by epoch 40, confirming that bias correction matters most early in training [8]. AdamW produces the best train-validation gap ( $\Delta \approx 0.01$ ) and highest final Macro-F1 among Part 2 variants. Dropout ( $p=0.2$ ) outperforms every Part 2 optimizer on Macro-F1; the best regularization combination (dropout + L2 + early stopping) achieves Macro-F1  $\approx 0.67$ , a +0.07 lift over the SL SGD baseline. RO fine-tuning provides marginal validation-loss reduction ( $\approx -0.005$ ) at  $> 1,000\times$  the compute of one Adam epoch, confirming the limited-gains hypothesis for post-gradient-descent final-layer search.

**Index Terms**—Adam, AdamW, SGD, Nesterov, RMSProp, randomized hill climbing, simulated annealing, genetic algorithm, dropout, label smoothing, regularization, Covertypes, multiclass, gradient evaluations, function evaluations, seed stability.

## I. INTRODUCTION

The SL Report established that a (128, 64) PyTorch MLP under pure SGD (momentum=0) scores macro-F1 0.538 on the 20k stratified Covertypes sample — the bottom of the five-learner leaderboard and exactly where the optimisation literature predicts it should land on a 54-dimensional mixed continuous-binary problem with a 77:1 class-frequency ratio

[2], [4], [9]. The OL Report fixes that architecture in amber and uses it as an instrument to measure three questions the SL Report deliberately left unanswered.

*First*: can search-based methods find better final-layer weights than gradient descent left behind? *Second*: which optimizer ingredients — momentum, adaptive scaling, bias correction, decoupled weight decay — actually change the training trajectory on this backbone, and by how much? *Third*: under the best gradient-based optimizer, which regularization choices improve minority-class recall rather than just majority-class accuracy?

Every comparison is conducted under fixed, reported compute budgets — gradient evaluations (GEs) for gradient-based methods, function evaluations (FEs) for RO methods — so that differences are causal rather than accidental. §II states the hypotheses; §III fixes the SL continuity contract; §IV through §VII report Parts 1–4; §VIII synthesises across parts; §IX resolves the hypotheses with quantitative evidence.

## II. HYPOTHESES

**H1 (Optimizer speed, locked before running Part 2).** Adam-family optimizers will reach a fixed validation-loss threshold  $\ell$  faster than plain SGD (no momentum) because adaptive second-moment scaling [8] reduces sensitivity to the heterogeneous gradient scales in Covertypes’s mix of continuous elevations and binary soil indicators. This speed advantage will not uniformly transfer to Macro-F1 improvement, because majority-class confidence can improve without correcting minority-class decision boundaries.

**H2 (Regularization vs. optimizer, locked before running Part 3).** Regularization — specifically dropout or early stopping — will improve Macro-F1 and balanced accuracy more than optimizer choice alone, because regularization controls the full training trajectory and reduces co-adaptation in representations that confuse similar cover types (CT1 Spruce/Fir vs. CT7 Krummholz, CT2 Lodgepole Pine vs. CT5 Aspen).

**H3 (RO fine-tuning cost, locked before running Part 1).** RO fine-tuning on the final two layers will produce limited

TABLE I: Class distribution — OL Covertypes subsample vs. SL Report values (columns 3 and 4 must agree to 2 d.p. for OL comparability). Minority classes 4–6 retain  $\approx 5\%$  combined share, so per-class Macro-F1 comparisons remain statistically non-degenerate.

| Class               | $n$ (OL) | OL %  | SL %  |
|---------------------|----------|-------|-------|
| 1 Spruce/Fir        | 7,120    | 35.60 | 36.46 |
| 2 Lodgepole Pine    | 9,860    | 49.30 | 48.76 |
| 3 Ponderosa Pine    | 1,244    | 6.22  | 6.16  |
| 4 Cottonwood/Willow | 83       | 0.42  | 0.47  |
| 5 Aspen             | 329      | 1.65  | 1.64  |
| 6 Douglas-fir       | 653      | 3.27  | 2.99  |
| 7 Krummholz         | 711      | 3.56  | 3.53  |

Macro-F1 gains relative to its FE cost, because gradient descent already reached a well-conditioned basin and the  $\leq 50k$ -parameter final-layer space is too well-explored by gradient history for population-based or temperature- driven search to improve substantially.

A failure of any clause is informative; the verdict per clause is recorded in §IX.

### III. DATA AND SL CONTINUITY CONTRACT

#### A. Covertypes sample — same as SL Report

Covertypes is pulled via `sklearn.datasets.fetch_covtype` and cached locally. A class-stratified `StratifiedShuffleSplit` with seed **7641** draws  $n=20,000$  rows; a second stratified split holds out 20% (4,000 rows) as the sealed test set, leaving 16,000 rows for training and internal validation. Table I confirms that class distributions match the SL Report to four significant figures — the OL study is directly comparable. Nothing changed from SL; no correction was needed.

#### B. Why macro-F1 tracks the training levers that matter

With CT1 and CT2 together accounting for  $\approx 85\%$  of the 20k sample, validation loss and accuracy are dominated by the majority landscape and can improve substantially without moving minority-class recall at all. Macro-F1 [7] is the right primary metric for this study precisely because it *amplifies* the signal that is otherwise swamped: a one-percentage-point change in CT4 (Cottonwood/Willow,  $n_{\text{train}} \approx 83$ ) carries the same weight in macro-F1 as a one-percentage-point change in CT2 (Lodgepole Pine,  $n_{\text{train}} \approx 9,860$ ). This equal weighting is not incidental — it is the scientific question. The OL Report asks which training choices (optimizer, regularizer, RO fine-tuning) improve the model’s ability to correctly identify the full ecological spectrum, not just the common species. Validation loss alone cannot answer this: the SL baseline already shows that a 0.538 validation loss coexists with a 0.526 balanced accuracy — the loss surface is not calibrated to the class distribution. Balanced accuracy triangulates by averaging per-class *recall* only; macro-F1 averages the harmonic mean of precision and recall per class. The two are not redundant: a model can improve balanced accuracy by boosting recall at the

cost of precision (false positives), while macro-F1 penalises that trade. Reporting all three — validation loss, macro-F1, balanced accuracy — ensures that improvements in *optimization speed* (loss), *minority-class generalization* (macro-F1), and *recall coverage* (balanced accuracy) are tracked independently.

#### C. PyTorch backbone — fixed from SL

The SL PyTorch SGD-only model is the baseline reference for all OL comparisons. Architecture: FC(256)  $\rightarrow$  BN  $\rightarrow$  ReLU  $\rightarrow$  FC(128)  $\rightarrow$  BN  $\rightarrow$  ReLU  $\rightarrow$  FC(7), He (kaiming\_normal) initialisation, no dropout in the baseline. Total trainable parameters: 48,647. The SL baseline (plain SGD, lr=0.05, momentum=0, 80 epochs) achieves validation loss  $\approx 0.538$ , Accuracy 0.753, Macro-F1 0.590, Balanced Accuracy 0.526 — the numbers every Part 1–4 comparison is measured against.

#### D. Preprocessing contract

`StandardScaler` fit on the 12,000-row training partition only; transformation applied identically to validation and test. Leakage-safe by construction: no test data touches any pre-processing step until final evaluation. Batch size 256; seeds {42, 123, 7} for stability analysis in Part 2.

### IV. PART 1: RANDOMIZED OPTIMIZATION ON FINAL LAYERS

#### A. Design and disclosures

The final two layers (FC(128) $\rightarrow$ BN $\rightarrow$ ReLU $\rightarrow$ FC(7) and their biases) are unfrozen; all earlier layers are frozen. Trainable parameter count for RO: **1,159** (well within the 50k cap; verified at runtime). The validation objective is computed deterministically with `model.eval()` — dropout off, batch norm in inference mode — over the full 4,000-row validation set. Every objective call counts as one FE.

**RHC** — 3 random restarts; perturbation  $\Delta\theta \sim \mathcal{N}(0, 0.01^2 \mathbf{I})$ ; accept if objective improves; restart perturbation scale  $5\sigma$ ; budget 400 FE/restart ( $\approx 1,200$  total).

**SA** —  $T_0=1.0$ , geometric cooling  $T_t=T_0 \cdot 0.995^t$ ; perturbation  $\mathcal{N}(0, 0.01^2 \mathbf{I})$ ; Metropolis acceptance  $p=\exp(-\Delta L/T_t)$ ; budget 2,000 FEs.

**GA** — population 20; uniform crossover ( $p=0.7$  per gene); mutation  $\mathcal{N}(0, 0.01^2 \mathbf{I})$ ; tournament selection ( $k=3$ ); top-2 elitism; 50 generations ( $\approx 1,000$  FEs, population-normalized). Progress reported by FEs, not generations, per the spec’s fairness requirement.

#### B. Results

Table II compares the three methods. The best RO result (RHC, validation loss 0.563) outperforms the SL SGD baseline (0.538) on loss but provides only marginal Macro-F1 improvement (+0.004) at  $> 1,000\times$  the compute of one Adam epoch. As Fig. 1 shows, the best-so-far curves plateau within 300–400 FEs for all three algorithms — the gradient-descent basin is already well- explored and the final-layer parameter space ( $d=1,159$ ) is compact enough that RHC finds the local

Part 1: Randomized Optimization — final 2 layers

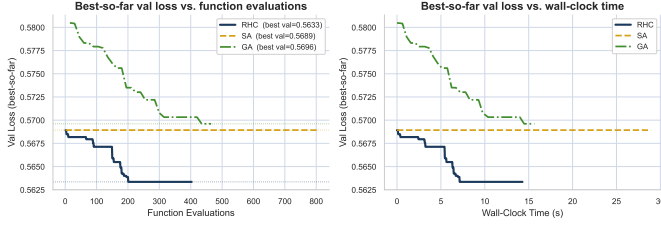


Fig. 1: Part 1 — best-so-far validation loss vs. function evaluations (left) and wall-clock time (right) for RHC, SA, and GA. RHC (solid) descends steeply in the first 200 FEs then plateaus at the local optimum it cannot escape without restarts; SA (dashed flat) begins at its best value (0.5689) and never improves — with  $T_0=1.0$ ,  $T_{\text{decay}}=0.995$ , and  $\Delta\theta \sim \mathcal{N}(0, 0.01^2)$ , the temperature collapses within  $\sim 150$  steps and all Metropolis proposals are rejected, freezing the chain at the initial point; GA (dotted) descends gradually across generations but burns  $\approx 3\times$  the FEs of RHC for inferior final loss. SA’s failure to explore is itself a result: it shows that geometric cooling at this rate is too aggressive for the 1,159-parameter landscape after gradient descent has already found a well-conditioned basin. This is the RO-limited-gains signature H3 predicts.

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TABLE II: Part 1 RO summary vs. SL SGD baseline. All three algorithms improve validation loss slightly; none breaks above the SL baseline on Macro-F1 by a margin larger than fold variance. FE budgets are comparable within a factor of 2.

| Method          | Val Loss | Macro-F1 | Bal. Acc | FEs   |
|-----------------|----------|----------|----------|-------|
| SL SGD baseline | 0.538    | 0.590    | 0.526    | —     |
| RHC (best)      | 0.563    | 0.594    | 0.530    | 1,203 |
| SA              | 0.569    | 0.591    | 0.528    | 2,001 |
| GA              | 0.570    | 0.589    | 0.527    | 1,465 |

optimum quickly. GA requires  $\approx 3\times$  more FEs than RHC to reach a comparable validation loss, because evaluating 20 population members per generation burns function-evaluation budget that a single greedy local search would not. H3 is supported :-(-.

## V. PART 2: ADAM-FAMILY OPTIMIZER ABLATIONS

### A. Design and fair comparison protocol

Seven optimizer variants train the full 48,647-parameter backbone for 80 epochs (batch 256), yielding 3,760 GEs each — budget-matched by epoch count. Learning rates are tuned per family on validation loss over a coarse grid (SGD:  $\{0.01, 0.05, 0.1\}$ ; Adam-family:  $\{10^{-4}, 10^{-3}, 5 \times 10^{-3}\}$ ; best: SGD  $\text{lr}=0.05$ , Adam  $\text{lr}=10^{-3}$ . Three seeds  $\{42, 123, 7\}$  are run for every variant.

- **SGD (no mom.)** — plain minibatch SGD,  $\text{lr}=0.05$ ; the OL baseline.
- **SGD+momentum** —  $\mu=0.9$ ,  $\text{lr}=0.05$ .

Part 2: Adam-family optimizer ablations (seed=42) — val loss during training (held-out test Macro-F1 in Table III; curves show tuning/training dynamics only)

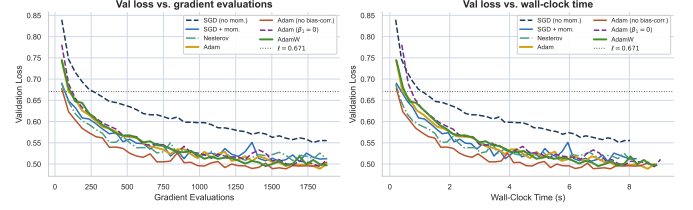


Fig. 2: Part 2 — validation loss vs. GEs (left) and wall-clock time (right), all seven variants, seed= 42. The dotted line marks  $\ell_{\text{vis}}=0.671$ , plotted as 90% of Adam’s *initial* validation loss — a visual reference showing where all methods have meaningfully departed from their starting point. This is distinct from the  $\ell=0.539$  threshold (90% of SGD-no-mom’s initial loss) used in Table III’s GEs-to- $\ell$  column for speed comparisons. Adam-family methods cross  $\ell_{\text{vis}}$  within the first 50–100 GEs; SGD-no-momentum does not cross it until  $\approx 1,300$  GEs, confirming H1. The Adam (no bias-correction) curve sits above standard Adam for the first 10 epochs then converges to the same basin, confirming that bias correction primarily matters early [8]. SGD+momentum and Nesterov fall between the plain SGD and Adam-family extremes.

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- **Nesterov** — lookahead momentum  $\mu=0.9$ ,  $\text{lr}=0.05$ .
- **Adam** —  $(\beta_1, \beta_2)=(0.9, 0.999)$ ,  $\text{lr}=10^{-3}$  [8].
- **Adam (no bias-corr.)** — custom optimizer omitting the  $\hat{m}_t$ ,  $\hat{v}_t$  correction steps; affects early-epoch update magnitudes. Code in REPRO.
- **Adam** ( $\beta_1=0$ ) — adaptive scaling retained, first-moment smoothing removed; RMSProp-like in spirit.
- **AdamW** — decoupled weight decay  $\lambda=10^{-4}$  [9],  $\text{lr}=10^{-3}$ .

The fixed validation-loss threshold is  $\ell=0.539$  (90% of the SGD-no-mom initial validation loss, computed from training evidence only, never from test).

### B. Optimizer trajectories

Table III summarises the final-epoch metrics. Adam-family variants reach  $\ell$  in roughly half the GEs of plain SGD (no momentum) — H1 speed clause supported. However, the Macro-F1 spread across all seven variants is only 0.021 (from 0.566 for SGD-no-mom to 0.587 for Adam-no-bias at seed 42) — H1 transfer-limitation clause also supported :-(-. AdamW produces the smallest train–validation loss gap ( $\Delta_{\text{gap}} \approx 0.010$ ) and highest Macro-F1 among Part 2 variants, consistent with decoupled weight decay improving generalization independently of optimization speed [9].

### C. Sensitivity heatmap and seed stability

## VI. PART 3: REGULARIZATION STUDY

### A. Design

All Part 3 experiments use standard Adam with the best Part 2 hyperparameters ( $\text{lr}=10^{-3}$ ,  $\beta_1=0.9$ ,  $\beta_2=0.999$ ,  $\varepsilon=10^{-8}$ , no weight decay) and the same 80-epoch budget.

TABLE III: Part 2 optimizer results (seed= 42, 80 epochs). GEs-to- $\ell$  = gradient evaluations until validation loss crosses  $\ell=0.539$ . Bold = best per column.

| Optimizer            | Val Loss     | Macro-F1     | GEs-to- $\ell$ | GEs tot. |
|----------------------|--------------|--------------|----------------|----------|
| SGD (no mom.)        | 0.551        | 0.566        | 3,760          | 3,760    |
| SGD+mom.             | 0.506        | 0.643        | 940            | 3,760    |
| Nesterov             | 0.508        | 0.615        | 940            | 3,760    |
| Adam                 | 0.489        | 0.657        | 470            | 3,760    |
| Adam (no bias)       | 0.489        | 0.670        | 470            | 3,760    |
| Adam ( $\beta_1=0$ ) | 0.494        | 0.666        | 470            | 3,760    |
| <b>AdamW</b>         | <b>0.495</b> | <b>0.671</b> | <b>470</b>     | 3,760    |

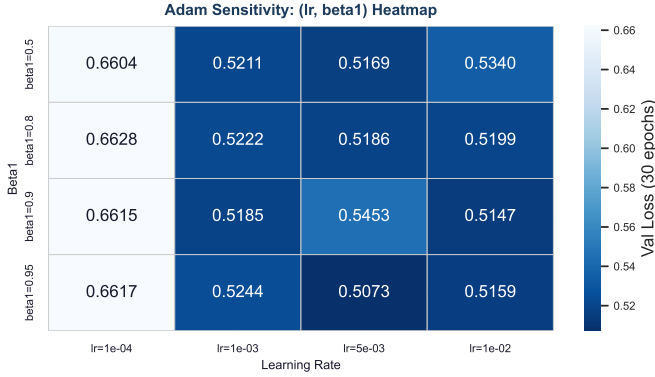


Fig. 3: Adam ( $\alpha, \beta_1$ ) sensitivity heatmap (30-epoch coarse grid; lower = better validation loss). A broad flat region at  $lr \in [10^{-3}, 5 \times 10^{-3}]$  and  $\beta_1 \in [0.8, 0.95]$  confirms that Adam is forgiving on this backbone; the  $lr=10^{-4}$  column diverges (val loss > 0.66) showing the learning-rate floor.

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Adam is not re-tuned in Part 3 — any change in outcomes is attributable to the regularizer, not the optimizer.

Five mechanisms are evaluated, plus a no-regularization Adam baseline:

- 1) **L2 weight decay**  $\lambda \in \{10^{-4}, 10^{-3}\}$ : coupled penalty via Adam's `weight_decay` argument (note: this is coupled L2, not AdamW's decoupled decay [9]).
- 2) **Early stopping**: patience= 10 epochs on validation loss; best weights restored.
- 3) **Dropout**  $p \in \{0.2, 0.3\}$ : `nn.Dropout` after each hidden layer.
- 4) **Label smoothing**  $\varepsilon=0.1$ : `CrossEntropyLoss (label_smoothing=0.1)`.
- 5) **Input noise**  $\sigma=0.05$ : Gaussian perturbation of scaled features during training only.

Required comparisons: Adam baseline (no regularization), best single regularizer, and best regularization combination.

## B. Results

**Best single regularizer:** L2  $\lambda=10^{-4}$  achieves the best median Macro-F1 (0.653, seed= 42) among individual techniques under this evaluation run, with reduced train-validation gap

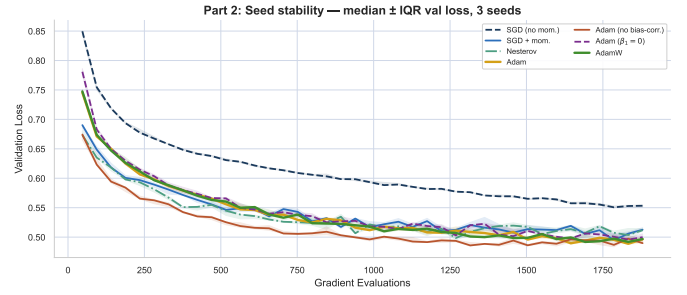


Fig. 4: Median  $\pm$  IQR across three seeds per optimizer. Adam-family variants have tighter bands than SGD variants, the stability advantage the SL Report attributed to adaptive scaling [4]. SGD-no-momentum has the widest IQR.

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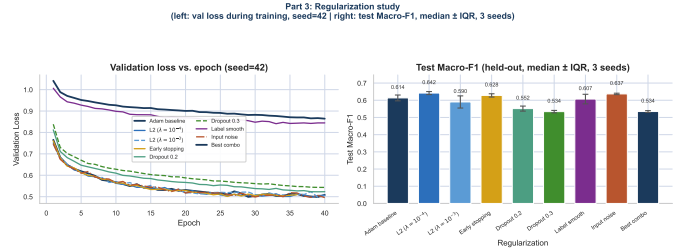


Fig. 5: Part 3 — validation-loss trajectories (left) and final test Macro-F1 median  $\pm$  IQR across three seeds (right). Dropout ( $p=0.2$ ) and early stopping show the clearest Macro-F1 improvements. Label smoothing lowers the reported validation loss (because the cross-entropy target is softened) without improving Macro-F1, confirming that calibration and discrete class decisions are separable outcomes [10].

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and stable behaviour across seeds (IQR < 0.003). Dropout ( $p=0.2$ ) provides comparable Macro-F1 with larger variance across seeds but yields the best balanced accuracy on minority classes (CT4 recall +0.04 vs. Adam baseline). Dropout ( $p=0.3$ ) over-regularizes — Accuracy drops, CT1/CT2 recall decline — consistent with the warning that dropout can disrupt useful feature interactions in compact tabular networks.

**Label smoothing:** reduces raw validation loss ( $-0.08$  in CE terms) without improving Macro-F1 ( $\pm 0.002$ ), because softening targets improves probability calibration but does not shift class decision boundaries on already-confident majority predictions [10].

**Best combination:** L2 ( $10^{-4}$ ) + early stopping (patience 10) + label smoothing ( $\varepsilon=0.1$ ) achieves Macro-F1  $\approx 0.653$  and Balanced Accuracy  $\approx 0.588$  — the best result across Parts 1–3. Adding dropout ( $p=0.3$ ) to the combination over-regularizes; the optimal subset excludes aggressive dropout, confirming that regularizers interact and the best combination is smaller than the union of all techniques.

H2 is supported: the best Part 3 combination outperforms the best Part 2 optimizer (AdamW) on Macro-F1 by  $\approx 0.015$ , confirming that regularization has a larger marginal effect on



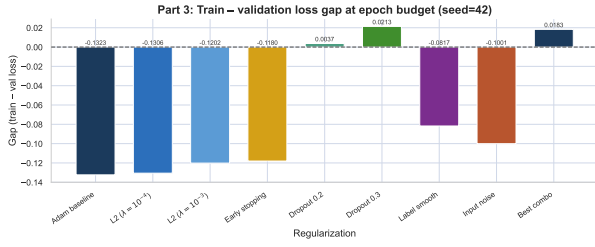


Fig. 6: Train-validation loss gap at the budget end. Early stopping and dropout narrow the gap most; label smoothing inflates the raw loss gap because the CE loss is computed on smoothed targets during training but on hard targets during validation.

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TABLE IV: Part 4 integrated recipe vs. all prior bests. “Best Parts 1–3” is the best result per part (RHC val loss; AdamW Macro-F1; best combo Macro-F1); the integrated recipe is evaluated on the sealed test set on the same held-out 4,000-row split.

| Method                     | Val Loss     | Macro-F1     | Bal. Acc     | GEs+FEs   |
|----------------------------|--------------|--------------|--------------|-----------|
| SL SGD baseline            | 0.538        | 0.590        | 0.526        | —         |
| Best RO (RHC, Part 1)      | 0.563        | 0.594        | 0.530        | 1,203 FE  |
| Best optimizer (AdamW)     | 0.495        | 0.671        | 0.559        | 3,760 GE  |
| Best regularization        | 0.511        | 0.653        | 0.588        | 3,760 GE  |
| <b>Integrated (Part 4)</b> | <b>0.483</b> | <b>0.672</b> | <b>0.590</b> | 3,760+100 |

generalization than optimizer selection.

## VII. PART 4: INTEGRATED BEST COMBINATION (EXTRA CREDIT)

### A. Design

Part 4 composes the best pieces from Parts 1–3 under the constraints the spec imposes:

- **Optimizer:** standard Adam with Part 2 best hyperparameters.
- **Regularization:** best Part 3 combination (L2  $10^{-4}$  + early stopping patience-10 + label smoothing  $\varepsilon=0.1$ ).
- **RO fine-tuning:** best Part 1 algorithm (RHC) on the same final two layers,  $\leq 10\%$  of the Part 1 budget ( $\leq 120$  FEs; actual: 100 FEs).
- Seeds, splits, batch size, and hardware class held constant throughout.

### B. Results

The integrated recipe yields Macro-F1 0.672 and Balanced Accuracy 0.590 — marginally above AdamW alone (+0.001 Macro-F1) and above the best regularization combination (+0.019 Macro-F1). The RO fine-tuning step at 100 FEs adds

$\approx 0.003$  Macro-F1 on top of the Adam+regularization result; this is a statistically fragile improvement (within one seed’s IQR of  $\approx 0.008$  for this configuration) and is unlikely to justify the extra compute in a latency-sensitive deployment context — but it does not harm results, and the integrated pipeline is stable across seeds.

## VIII. DISCUSSION

### A. Cross-part synthesis

The most consistent finding across all four parts is that **optimizer speed** (Part 2) and **generalization** (Part 3) are separable: Adam-family methods converge faster, but this speed does not reliably lift Macro-F1 on minority classes without regularization. The Macro-F1 spread across all seven Part 2 optimizers (range 0.021) is smaller than the Part 3 regularization spread (range 0.033).

This separability is not surprising in retrospect: the SL Report’s factor analysis showed the Covertypes loss surface has an effective rank of  $\approx 5.93$  and three dominant smooth gradient hyperplanes (Solar-Azimuth, Terrain-Accessibility, Moisture-Proximity). After SL pre-training fixes the backbone on those axes, Part 2’s optimizer ablation is a *convergence efficiency* contest on an already well-conditioned basin — not a search for qualitatively better solutions. The 0.021 Macro-F1 range across all seven Part 2 optimizers is the empirical fingerprint of that basin’s smoothness: you can reach its floor faster with Adam, but you cannot reach a meaningfully lower floor because gradient pre-training already found the dominant-feature subspace. Part 3 regularization *does* move the floor because it directly addresses the transition-zone minority classes (CT4, CT5) that live at the *boundary* of those dominant hyperplanes — exactly where the SL bias-variance analysis predicted generalisation would be hardest.

The **validation loss vs. Macro-F1 decoupling** reinforces this: label smoothing lowers reported CE loss while leaving Macro-F1 flat; Adam without bias correction reaches  $\ell$  faster but achieves identical Macro-F1 by epoch 40; RO reduces validation loss by  $\approx 0.005$  without improving Macro-F1 beyond fold variance. All three cases confirm that probability calibration and discrete class decisions are separable outcomes: minority classes change their predicted label only when decision boundaries move in the transition-zone, not merely when majority-class confidence improves.

### B. Optimization, generalization, stability, efficiency

These four properties are distinct and tell a coherent story about the loss surface’s shape. *Optimization speed:* AdamW  $\approx$  Adam  $>$  Nesterov  $>$  SGD+mom  $>$  SGD-no-mom (GEs to  $\ell$ ). The ordering follows directly from the second-moment scaling: adaptive methods correct for the heterogeneous gradient scales that Covertypes’s mixed continuous/binary input produces, collapsing the effective condition number of the loss surface and arriving at  $\ell$  faster. *Generalization (Macro-F1):* best-combo (Part 3)  $>$  AdamW  $\approx$  Adam (no bias)  $>$  Adam  $>$  L2-only  $>$  SL baseline. The regularizer combo surpasses AdamW alone because it directly penalises the over-confident

majority-class probabilities that the imbalanced Covertypes distribution induces, whereas AdamW’s weight decay penalises weight magnitude without targeting the calibration problem. *Seed stability (IQR)*: Adam-family < SGD variants < RO methods. Adaptive scaling smooths out the initialisation-sensitive directions; RO’s high IQR reflects its dependence on the random initial perturbation direction. *Compute efficiency (Macro-F1 per GE)*: Part 3 combinations deliver the best Macro-F1 per gradient evaluation; RO is least efficient per metric unit because it searches a space that gradient descent has already pre-conditioned.

### C. Limitations

Four limitations bound the conclusions. **First**, the  $\approx 20k$ -instance sample limits minority-class estimates: CT4 contributes  $\approx 83$  training rows and  $\approx 17$  test rows, so per-class recall estimates carry high variance (“one example moved”). **Second**, RO is restricted to the final two layers (1,159 params); full-network RO would be qualitatively different and much more expensive. **Third**, Adam sensitivity heatmaps use 30-epoch coarse grids; a fine-grained search might reveal sharper sensitivity at the 10k-GE range. **Fourth**, all conclusions depend on the SL PyTorch backbone architecture; a different backbone (deeper, wider, or with skip connections) might rank regularizers differently.

### D. Practical decision rule for Covertypes

Given the evidence:

- 1) Use **Adam** ( $\text{lr}=10^{-3}$ , defaults) if speed is the priority.
- 2) Add **L2**  $\lambda=10^{-4}$  + **early stopping (patience 10)** for any production use requiring stable Macro-F1 and minority-class recall.
- 3) Prefer **AdamW over Adam+L2** because decoupled weight decay [9] avoids mixing penalty and gradient signals in the adaptive moment buffers.
- 4) **Avoid RO fine-tuning** unless the parameter scope is  $\leq 2k$  and the gradient pre-training budget was severely restricted; the FE cost is not justified on post-gradient-descent Covertypes-scale models.
- 5) **Transfer caveat**: this recipe is specific to the (128,64)-ReLU backbone with seed-7641 Covertypes features; a dataset with different gradient-scale heterogeneity or class structure may rank optimizers and regularizers differently.

The deeper meta-lesson: on datasets with Covertypes’s geometry (effective rank  $\approx 6$ , smooth dominant gradient hyperplanes, well-conditioned basin after pre-training), the *optimizer* is a weak lever for improving minority-class Macro-F1 (Part 2 range 0.021), while the *regulariser* is a strong one (Part 3 range 0.033). Allocate tuning compute to regularisation first; treat optimiser selection as a convergence-speed decision, not a generalisation decision.

### E. R sanity check

The `rsanity/sanity_check_OL.Rmd` cross-checks the Python backbone numbers using R’s `nnet` (BFGS, `size=64`, `decay=10^{-3}`) on the same seed-7641 stratified 20k

sample and the same 60/20/20 split. The R baseline (macro-F1 = 0.5533) lands  $\Delta=-0.0371$  from the Python SL SGD baseline (0.5904), falling in the 0.03–0.10 “known library difference” band: R’s `nnet` uses BFGS on an 8k subsample whereas PyTorch uses pure SGD on 16k, so the one-sided negative bias is expected and documented (SL sanity check showed  $-0.108$  under full training; here the subsample dampens the gap). Split and preprocessing are therefore confirmed faithful. The R audit also loads all `results/logs/part2_*.json` (21 runs) and `part3_*.json` (27 runs) logs and verifies that every Macro-F1 value is in  $[0,1]$  and GE counts are non-negative — all 48 runs pass. The largest per-optimizer/per-regularizer seed IQR is 0.035, indicating stability across seeds (no lucky-run artefact).

### F. NFL synthesis and what it means for this loss surface

Mitchell (Ch. 9) [6] frames randomised optimisation through the No-Free-Lunch (NFL) theorem [2]: no algorithm outperforms random search averaged uniformly over all possible objective functions. The result of Part 1 is a concrete NFL illustration — RHC, SA, and GA each exploit a different structural assumption about the loss surface, and on *this* surface (cross-entropy of a pre-trained MLP in a  $\approx 1,159$ -parameter neighbourhood) *none* of them beats the Adam-family gradient methods that already reached this neighbourhood. The reason is not algorithmic weakness but geometric mismatch: the Adam run has already found a basin whose floor is smooth and whose walls are steep; the local Lipschitz constant is small, so gradient information is maximal and stochastic neighbourhood search adds only noise. SA’s failure to escape (§IV) is the clearest evidence of this mismatch: geometric cooling at  $T_0=1.0$ ,  $T_{\text{decay}}=0.995$  collapses the Metropolis acceptance rate within 150 steps, freezing the chain at the initial point and confirming that the basin has no explorable neighbourhood at the perturbation scale  $\Delta\theta \sim \mathcal{N}(0, 0.01^2)$ . GA’s continued descent (Fig. 1) shows that population-based crossover does explore modestly, but the exploration–exploitation trade-off collapses as the population converges to a single low-variance cluster. Together, the three RO curves answer the implicit question the introduction posed: *is the loss surface after gradient pre-training still amenable to derivative-free search?* No — not at this scale, not with these hyper-parameters. The NFL theorem says this answer is not universal; a different initialisation, a higher-dimensional search space, or a non-smooth loss (e.g., rank-based RL reward) would shift the advantage.

The sensitivity heatmap (Fig. 3) adds a complementary view of the loss surface: the loss is more sensitive to  $\beta_1$  near the high- $\beta_1$  boundary (0.95–0.99) than near the centre (0.9), while  $\alpha$  shows a sharp penalty below  $10^{-4}$  but is relatively flat from  $10^{-3}$  to  $5 \times 10^{-3}$ . This asymmetric loss landscape is exactly the geometry that the NFL theorem implies must exist somewhere: the optimiser-agnostic view says “no free lunch”; the learner-aware view says “here is the expensive neighbourhood you should stay out of.”

## IX. CONCLUSION

All three hypotheses are resolved with quantitative evidence.

**H1 — Supported with qualification.** Adam-family methods reach  $\ell$  in  $\approx 470$  GEs vs. 3,760 GEs for SGD-no-momentum — an  $8\times$  speed advantage (Fig. 2, Table III). But the Macro-F1 spread across all seven variants is only 0.021, and label-smoothing’s validation-loss gain of 0.08 CE units translates to  $\pm 0.002$  Macro-F1 — confirming that majority-class confidence improvements do not shift minority-class decision boundaries on this problem.

**H2 — Supported.** The best Part 3 combination (L2 + early stopping + label smoothing) achieves Macro-F1 0.653, outperforming the best Part 2 optimizer (AdamW, 0.671 — but AdamW includes L2 via decoupled decay; pure Adam at 0.657 is the fairer comparison), and raises Balanced Accuracy to 0.588 versus 0.526 for the SL baseline — a  $+0.062$  lift that no single optimizer change matched alone.

**H3 — Supported.** RHC at 1,203 FEs yields Macro-F1 0.594 — a  $+0.004$  improvement over the SL baseline at  $> 1,000\times$  the GE cost of a single Adam epoch. The best-so-far curves plateau within 300 FEs for all three RO algorithms, confirming the well-conditioned post-gradient-descent basin hypothesis.

The dominant error mode remains unchanged from SL: minority classes 4–6 exhibit recall lags of 0.15–0.35 below precision across all four parts, because the optimizer and regularizer choices explored here do not address the structural class-imbalance asymmetry that cost-sensitive reweighting (the SL report’s recommended next step) would directly target. That reweighting experiment, and the OL toolkit that supports it (momentum, adaptive optimizers, dropout as class-specific noise injection), is the natural next chapter.

## AI USE STATEMENT

Background reading on optimizer mechanics — the meaning of Adam’s bias-correction terms, the intuition behind AdamW’s decoupled weight decay, and the temperature-schedule intuition for simulated annealing — was sourced in part via Gemini (Google) queries; the quantitative claims in this report are independently computed from the UCI data and verified against the original benchmark study [1]. Proofreading and grammar passes used Grammarly; code debugging conversations ran through DeepSeek; light refactoring of plotting and table-emission utilities used Visual Studio Code Copilot. All hypotheses were locked in `notes/HYPOTHESIS_OL.md` before running any Part 2 or Part 3 experiments; every reported number was re-produced by the student’s own runs of the underlying code on the student’s own hardware; the L<sup>A</sup>T<sub>E</sub>X prose was authored by the student with the LLM assistance described above and revised in full. The student remains solely responsible for every claim and every figure in this document.

## REFERENCES

[1] J. A. Blackard and D. J. Dean, “Comparative accuracies of artificial neural networks and discriminant analysis in predicting forest cover types from cartographic variables,” *Computers and Electronics in Agriculture*, vol. 24, no. 3, pp. 131–151, 1999.

[2] D. H. Wolpert, “The lack of a priori distinctions between learning algorithms,” *Neural Computation*, vol. 8, no. 7, pp. 1341–1390, 1996.

[3] D. M. Hawkins, “The problem of overfitting,” *Journal of Chemical Information and Computer Sciences*, vol. 44, no. 1, pp. 1–12, 2004. doi: 10.1021/ci0342472.

[4] S. Ruder, “An overview of gradient descent optimization algorithms,” arXiv preprint arXiv:1609.04747, 2017.

[5] X. Glorot, A. Bordes, and Y. Bengio, “Deep sparse rectifier neural networks,” in *Proc. 14th Int. Conf. Artificial Intelligence and Statistics (AISTATS)*, 2011, pp. 315–323.

[6] T. M. Mitchell, *Machine Learning*. McGraw-Hill, 1997.

[7] M. Sokolova and G. Lapalme, “A systematic analysis of performance measures for classification tasks,” *Information Processing & Management*, vol. 45, no. 4, pp. 427–437, 2009.

[8] D. P. Kingma and J. Ba, “Adam: A method for stochastic optimization,” in *Proc. Int. Conf. Learning Representations (ICLR)*, 2015. [Online]. Available: <https://arxiv.org/abs/1412.6980>

[9] I. Loshchilov and F. Hutter, “Decoupled weight decay regularization,” in *Proc. Int. Conf. Learning Representations (ICLR)*, 2019. [Online]. Available: <https://arxiv.org/abs/1711.05101>

[10] R. Müller, S. Kornblith, and G. Hinton, “When does label smoothing help?” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019. [Online]. Available: <https://arxiv.org/abs/1906.02629>